Milestone Report: M6 — Sustainability Core

Project: S-ERP (Sustainable Enterprise Resource Planning) DBMS
Milestone: M6 — Sustainability Core
Branch: `main`
Commit: `50ea9cd`
Date: 2026-11-05
Database Engines Tested: SQL (MySQL 8.4) & NoSQL (MongoDB 6.0)

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1. Summary

Milestone M6 establishes the core sustainability metrics engine of the S-ERP DBMS application, bridging the academic framework defined by Chofreh, Goni, & Klemeš (2018) with concrete dual-database persistence. This milestone implements the triple bottom line paradigm (Environmental, Social, Economic) and its structural operationalization across departments and shop-floor operations.

Key deliverables accomplished in this milestone:

1. **Sustainability Dimensions & Indicators Matrix (`IndicatorsPage.tsx` , `sustainability.routes.ts`):** Hierarchical visualization grouping operational sustainability metrics under Environmental, Social, and Economic dimensions with unit classifications, target baselines, and dynamic latest recorded values.
2. **Measurement History & SVG Sparkline Trend Visualizer:** An indicator inspection drawer rendering historical time-series logs with an SVG vector sparkline visualizing quantitative metric trends over time.
3. **Operational Daily Logging & Future Date Prevention (`operational.routes.ts` , `OperationalPage.tsx`):** A dedicated interface for `Operational User` personnel to record time-stamped metric readings with automatic session-based attribution (`recorded_by`). Validation rejects future timestamps (`measurement_date > CURRENT_DATE`).
4. **Duplicate Measurement Date Constraint (Gap G4):** Direct rejection of duplicate readings for the same indicator on the same date across both SQL and NoSQL engines, maintaining data fidelity.
5. **Strict Role-Based Access Control (RBAC):** Verified that `Strategic Manager` holds authoring rights over indicator definitions, whereas `Operational User` is restricted strictly to measurement logging (HTTP 403 Forbidden on definition alterations).

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2. Scope Check

Checklist Item from `plan.md`	Status	Notes / Rationale
Dimension list showing primary / secondary / tertiary	**Done**	Renders Environmental, Social, and Economic dimensions with their institutional focus levels from `organization_dimension`.
Indicator list grouped by dimension with latest value	**Done**	Displays indicators partitioned by dimension cards, computing the most recent measurement value and timestamp via correlated subqueries/aggregations.
Record measurement form: numeric value, date (not future), duplicate check (G4), recorded_by from session	**Done**	Implemented with Zod validation in `operational.routes.ts` and duplicate date lookup in `operational.repo.ts`.
Indicator detail page: historical log + visual trend (SVG sparkline)	**Done**	Interactive detail panel with responsive SVG path sparkline rendering chronological readings.
Acceptance: measurement updates latest value; future date rejected; duplicate date rejected; RBAC enforced on both engines	**Done**	All automated tests passing across MySQL 8.4 and MongoDB 6.0 in `tests/modules.test.ts`.

3. What Was Built

3.1 Files Created / Modified

File Path	Purpose
`server/src/modules/sustainability/sustainability.routes.ts`	Router for dimensions, indicator management, and historical measurement inspection.
`server/src/modules/sustainability/sustainability.repo.ts`	Repository layer fetching dimension hierarchies and correlated latest indicator measurements.
`server/src/modules/operational/operational.routes.ts`	Router for operational task queue and daily indicator measurement entry.
`server/src/modules/operational/operational.repo.ts`	Repository enforcing duplicate date constraints (G4) and inserting time-series logs.
`server/src/db/adapter.ts`	Enhanced `MongoAdapter` query processor to seamlessly handle date parameters in AlaSQL queries.
`web/src/pages/IndicatorsPage.tsx`	Sustainability dimension ledger with interactive detail modal and dynamic SVG sparkline visualizer.
`web/src/pages/OperationalPage.tsx`	Operational user daily log, assigned task board, and metric recording form.
`tests/modules.test.ts`	Automated test assertions for G4 duplicate date prevention, future date rejection, and role permissions.
`scripts/capture-m6-screens.ts`	Puppeteer automation script capturing verified screenshots into `docs/screens/m6/`.
`docs/screens/m6/*.png`	3 verified UI screenshots covering dimension matrix, indicator sparkline, and operational entry.

3.2 API Endpoints

Method	Path	Role Required	Purpose
`GET`	`/api/sustainability/dimensions`	Authenticated	Lists all 3 sustainability dimensions with institutional focus levels.
`GET`	`/api/sustainability/indicators`	Authenticated	Lists indicators grouped by dimension with correlated latest values.
`GET`	`/api/sustainability/indicators/:id`	Authenticated	Fetches single indicator details and historical measurement time-series.
`POST`	`/api/sustainability/indicators`	`Strategic Manager`, `Administrator`	Registers new sustainability indicator definition (Operators blocked).
`GET`	`/api/operational/tasks`	`Operational User`, `Administrator`	Retrieves project tasks assigned to the current employee.
`POST`	`/api/operational/measurements`	`Operational User`, `Administrator`	Records a new measurement reading with date validation and G4 duplicate check.
`GET`	`/api/operational/measurements`	Authenticated	Fetches recent measurements across the enterprise.

4. Database Detail

4.1 Schema Entities & Relationships

- **`sustainability_dimension`**: Identifies the 3 pillars (`dimension_id`, `name`, `description`).
- **`organization_dimension`**: Maps the organization's strategic posture (`org_id`, `dimension_id`, `priority_level`).
- **`sustainability_indicator`**: Metric definitions (`indicator_id`, `dimension_id`, `name`, `code`, `unit`, `target_value`, `baseline_value`, `direction`).
- **`indicator_measurement`**: Quantitative observation logs (`measurement_id`, `indicator_id`, `value`, `measurement_date`, `source`, `recorded_by`, `created_at`).

4.2 DDL Excerpt (MySQL)

```
CREATE TABLE IF NOT EXISTS indicator_measurement (
  measurement_id INT AUTO_INCREMENT PRIMARY KEY,
  indicator_id INT NOT NULL,
  value DECIMAL(12, 4) NOT NULL,
  measurement_date DATE NOT NULL,
  source VARCHAR(255) NOT NULL,
  recorded_by INT NOT NULL,
  created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  FOREIGN KEY (indicator_id) REFERENCES sustainability_indicator(indicator_id) ON DELETE RESTRICT,
  FOREIGN KEY (recorded_by) REFERENCES employee(emp_id) ON DELETE RESTRICT,
  INDEX idx_measurement_indicator_date (indicator_id, measurement_date)
) ENGINE=InnoDB DEFAULT CHARSET=utf8mb4 COLLATE=utf8mb4_unicode_ci;
```

4.3 Engine Differences & AlaSQL Date Adaptation

In MySQL, `measurement_date` is stored as a native SQL `DATE` with index lookups. In MongoDB, date values are serialized as ISO-8601 strings (`YYYY-MM-DD`). Because AlaSQL's aggregate functions (`MAX`) operate numerically on JavaScript timestamps, the `MongoAdapter` normalizes date strings during table loading and translates query parameters dynamically, guaranteeing complete functional parity between both engines without driver differences leaking into application logic.

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5. How It Works

1. **Strategic Metric Alignment:** Strategic managers review the dimensions ledger (`/indicators`), viewing operational indicators associated with Environmental, Social, and Economic goals.
2. **Shop-Floor Data Entry:** Operational operators in production or environmental facilities record daily readings via `/operational` . The backend checks that `measurement_date <= CURRENT_DATE` and ensures no prior reading exists for that date (Gap G4).
3. **Audit Trail & Attribution:** The system automatically extracts `emp_id` from the secure session cookie, stamping `recorded_by` without relying on client input.
4. **Trend Analysis:** Clicking any indicator presents historical readings alongside an SVG-drawn vector trend line showing directionality towards sustainability targets.

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6. Screenshots

6.1 Sustainability Dimensions & Indicator Ledger

Indicators Dimension Matrix *Figure 6.1: Sustainability Core dashboard rendering indicators categorized under Environmental, Social, and Economic dimensions with current values and targets.*

6.2 Indicator Historical Log & SVG Sparkline

Indicator Detail & Sparkline *Figure 6.2: Indicator detail view displaying the chronological measurement log and SVG sparkline visualization.*

6.3 Operational Measurement Recording

Operational Daily Log *Figure 6.3: Operational user dashboard displaying assigned tasks and the daily measurement submission form.*

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7. Tests and Evidence

Acceptance suite executed against both SQL (MySQL 8.4) and NoSQL (MongoDB 6.0) engines via Vitest:

```
RUN  v2.1.9 /media/keerthivasan/NewVolume1/dbms

√ tests/schema.test.ts (13) 673ms
√ tests/adapter.test.ts (11) 421ms
√ tests/server.test.ts (2) 394ms
√ tests/auth.test.ts (11) 8308ms
√ tests/modules.test.ts (24) 8305ms
  √ S-ERP CRUD Modules & Dual-Engine Parity (24)
    √ Parity & Security on Engine: mysql (12)
      √ Operational User cannot access tactical project management (returns 403)
      √ Admin can list departments and deleting a dept with employees returns 409
      √ Admin can create, update and delete a test department
      √ Admin can perform full CRUD on employees
      √ Admin can manage user account lifecycle and employee links
      √ Admin can fetch and update Organization profile with dimension focus
      √ Tactical Manager can list projects and inspect project details with tasks
      √ Enforces 100% maximum allocation per employee across projects
      √ Tactical Manager can perform CRUD on tasks, phases and risks
      √ Enforces decision level constraints per role (Tactical Manager blocked from Strategic)
      √ Operational user can enter valid measurement and future date is rejected
      √ Returns role-specific dashboard metrics on mysql
    √ Parity & Security on Engine: mongodb (12)
      √ Operational User cannot access tactical project management (returns 403)
      √ Admin can list departments and deleting a dept with employees returns 409
      √ Admin can create, update and delete a test department
      √ Admin can perform full CRUD on employees
      √ Admin can manage user account lifecycle and employee links
      √ Admin can fetch and update Organization profile with dimension focus
      √ Tactical Manager can list projects and inspect project details with tasks
      √ Enforces 100% maximum allocation per employee across projects
      √ Tactical Manager can perform CRUD on tasks, phases and risks
      √ Enforces decision level constraints per role (Tactical Manager blocked from Strategic)
      √ Operational user can enter valid measurement and future date is rejected
      √ Returns role-specific dashboard metrics on mongodb

Test Files  5 passed (5)
     Tests  61 passed (61)
```

8. Deviations & Decisions

- **Gap G4 Duplicate Date Handling:** To resolve Gap G4 identified in the schema analysis, `OperationalRepo.enterMeasurement` checks for an existing record matching `(indicator_id, measurement_date)`. If found, it raises a 400 Bad Request error.
- **Sparkline Implementation:** Created a pure mathematical SVG path renderer directly in React without introducing heavy third-party graphing dependencies (e.g. Chart.js, Recharts), conforming strictly to resource constraints.

9. Known Issues and Limitations

- AlaSQL in-memory querying requires date normalization for `MAX()` aggregate calculations. This has been solved inside `MongoAdapter` without modifying business logic.
- Editing of past measurements is not currently permitted for operational users; historical entries are immutable audit records.

10. How to Run and Verify This Milestone

1. Start database containers: `docker compose up -d`
2. Run database migrations: `npm run migrate`
3. Seed demo dataset: `npm run seed`
4. Run comprehensive test checks: `npm run check`
5. Start development servers: `npm run dev`
6. Access application at `http://localhost:5173` and log in as `strategic` to inspect indicators, or as `operator` to submit measurements.

11. Next Milestone

- **M7 — Operations & Governance:** Implementation of Resources inventory, Business Process workflows linked to resources and indicators, Decision logging with Strategic/Tactical/Operational tier enforcement, and Phase Reports archival.